28 June 1968

MILITARY SPECIFICATION

PROPELLANT, FLUORINE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers the requirements for two types of fluorine propellant.

1.2 Classification. Fluorine propellant shall be of the following types, as specified:

Type I Gaseous

Type II Liquid

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-18455	Argon, Technical
MIL-P-27407	Helium

STANDARDS

Military

MIL-STD-101	Color Code for Pipelines and for Compressed-Gas Cylinders
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-172	Color Code for Containers of Liquid Propellants

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9135

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

Department of Transportation

49 CFR 171-190 Department of Transportation Hazardous Materials Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

MCA Manual L-1 A Guide for the Preparation of Warning Labels for Hazardous Chemicals (1961).

(Application for copies should be addressed to the Manufacturing Chemists' Association, Inc., 1825 Connecticut Avenue, N.W., Washington, D.C. 20009).

American Society for Testing and Materials Publications

ASTM Standards, Parts 17, 18, and 30.

(Copies of ASTM publications may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the propellant, Types I and II, shall conform to those listed in table I when tested in accordance with the applicable test methods.

TABLE I Chemical Composition

Composition	Limits (Percent by volume)	Test methods
Fluorine (F_2) Assay	99.0 min	4.5.2
Sum of Hydrogen Fluoride (HF) + Carbon Dioxide (CO_2)	0.1 max	4.5.3
Non-condensables*	0.9 max	4.5.1

*As defined in test method, e.g. O_2, N_2, He, CF_4.

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3.2 Filling pressure. Cylinders containing Type I fluorine propellant shall be filled as specified in 5.1.1.

3.3 Cylinder processing. Cylinder processing, including refilling and hydrostatic testing, shall be in accordance with approved contractors detailed procedures and shall be included in the contract or order.

3.3.1 Hydrostatic test data. Each cylinder shall be hydrostatically tested at intervals not to exceed 5 years in accordance with Hazardous Materials Regulations of the DoT, Paragraph 73.34. The test date shall be stamped on each cylinder.

3.4 Limiting values. The following applies to all specified limits in this specification: For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the propellant shall be classified as quality conformance tests.

4.3 Test conditions. The test conditions are described under the individual tests to which they apply.

4.4 Quality conformance tests. Quality conformance test shall consist of:

Individual tests 4.4.1

4.4.1 Individual tests. The propellant shall be subjected to tests as described in 4.5.

4.4.1.1 Sampling plan.

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4.4.1.1.1 Lot. A lot shall consist of propellants produced by one manufacturer and contained in one storage tank, provided the operation is continuous. In the event the process is a batch process, each batch shall constitute a lot.

4.4.1.1.2 Sample. A sample consists of 10 ± 2 liters of gaseous propellant at ambient conditions. Unless otherwise specified, quality conformance tests shall be made on the sample taken directly from the shipping containers or cylinders.

4.4.1.1.3 Cylinders. The number of cylinders selected from each lot shall be in accordance with MIL-STD-105, Inspection Level S-3. Each lot in a shipment shall be tested in accordance with table I.

4.4.1.1.4 Containers. When Type II fluorine propellant containers are filled from manufacturer's plant system or other storage facilities for delivery to destination by any mode of transportation, the sampling and acceptance tests shall be conducted on the container contents prior to leaving the site of filling.

4.4.1.2 Sample test preparation.

4.4.1.2.1 Fluorine manifold preparation. Vessels No. 11 to 17 inclusive, as shown in figure 1, shall be constructed with specific materials and shall contain specific reagents, the preparation of which is outlined as follows.

4.4.1.2.1.1 Manifold fabrication. Vessel No. 11 is the hydrofluoric acid absorber. It consists of a 3/8 X 24-inch long heavy wall copper tube, packed with sodium fluoride reagent, held in place with 80 mesh Monel screens. Each end is fitted with a copper pipe thread-to-1/4 inch Swaglok adapter. One-fourth inch copper tubing is used to connect vessel No. 11 to V_5 and vessel No. 12. Vessel No. 12 is the fluorine reactor and consists of a heavy wall 1.5-inch-diameter by 48-inch-long aluminum pipe fitted with threaded aluminum caps at each end. Vessel No. 12 is filled with prepared sodium chloride reagent which is held in place with 80 mesh Monel screens. One-fourth inch copper tubing is used to connect the top of vessel No. 12 to V_{19}. Vessel No. 13 is an inverted, calibrated, 500 ml dispensing buret fitted with an appropriately sized two-hole rubber stopper. Vessel No. 14 is a conventional leveling bulb, connected to vessel No. 13 with plastic tubing. Vessels No. 15 and 16 are 250 ml Fleming bottles used to scrub chlorine gas during the passivation of the manifold assembly shown in figure 1. S_7, S_8, and S_9 are three-way stopcocks connected to associated glassware with plastic tubing. The leveling bulb, vessel No. 14, is placed in a ring clamp attached to a ring stand in proximity to vessel No. 13. The stopper on vessel No. 13 must be wired to prevent accidental spillage.

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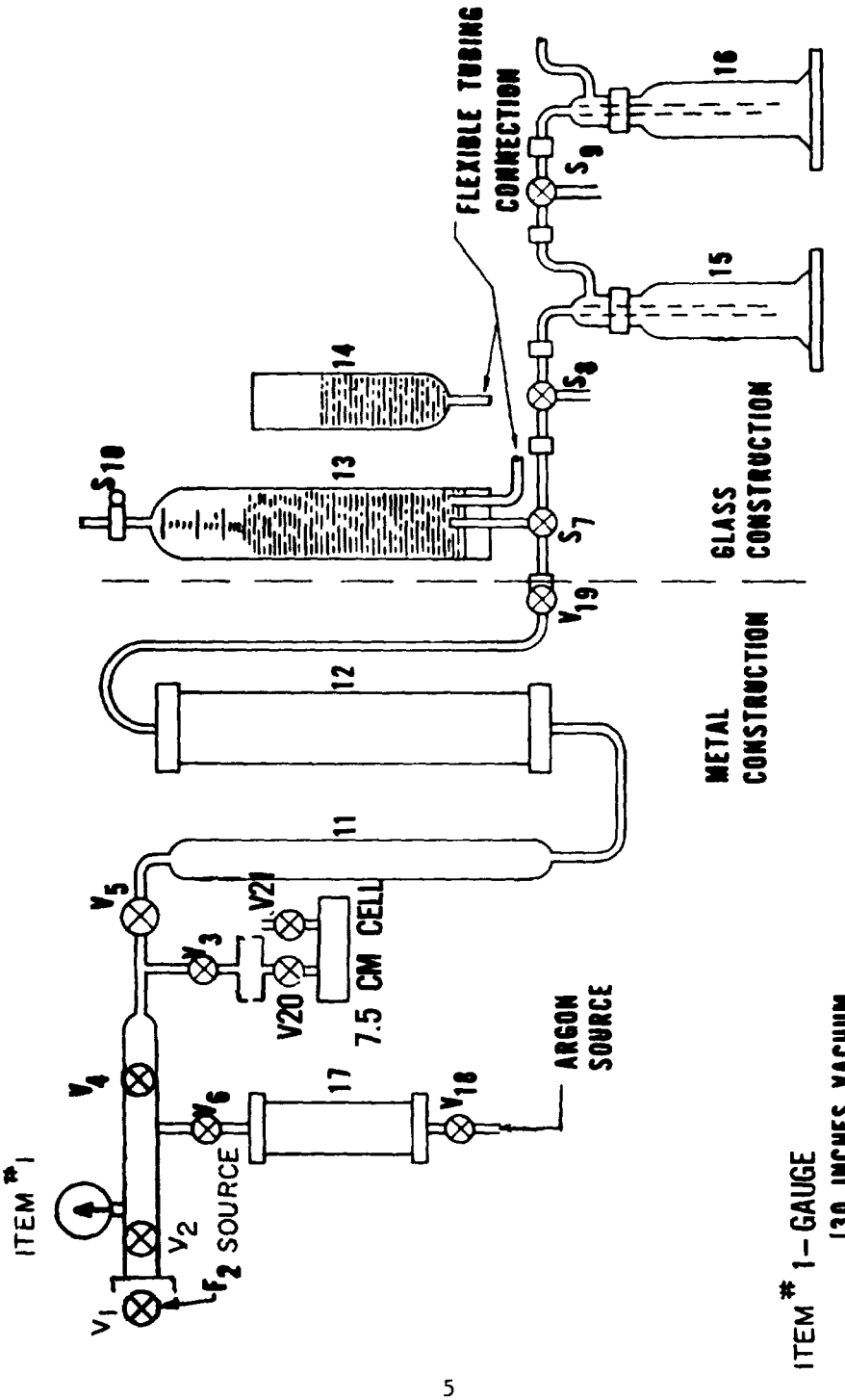


Figure 1. Fluorine Assay Manifold

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Vessel No. 17 is an argon drier and consists of 2-inch diameter iron pipe by 12 inches long with threaded caps, into which reducing nipples are screwed so that 1/4 inch tubing can be fitted with Swagelok fittings to connect the drier with V_{18} and V_6. V_6 is connected to a brass "T" on the manifold.

4.4.1.2.1.2 Reagents and equipment. The following reagents and equipment shall apply as conditions of 4.4.1.2.1.

(a) Reagents:

(1) Sodium fluoride, NaF, for the hydrogen fluoride reactor, No. 11: Heat NaF, ACS reagent grade, for 16 hours at 1472°F (800°C), cool, and screen to 10-40 mesh. Re-dry the 10-40 mesh NaF for 1 hour at 257°F (125°C) just prior to packing No. 11.

(2) Sodium chloride, NaCl, for the fluorine reactor, No. 12: Combine 5 lbs of rock salt (water softener grade) with 3 g of sodium fluoride in a large bottle, mixing the two reagents on a roller or tumbler. Pack No. 12, the aluminum tube, with the mixed salts.

(3) Alkaline arsenite solution, reactor No. 13, leveling bulb 14, and scrubbers No. 15 and 16: Dissolve, with cooling, 1154 g of ACS reagent grade potassium hydroxide, KOH, in 1800 ml of chloride-free distilled/deionized water. In a second erlenmeyer flask slurry 460 g of ACS reagent grade arsenic trioxide in 200 ml of chloride-free water. Add the slurry to the KOH solution, stir until a clear solution is obtained. No. 13 is filled through leveling bulb 14 until 50 ml free volume remains in No. 14. The Fleming bottles, 15 and 16, are filled three-fourths full.

(4) Phosphorus pentoxide/asbestos drying reagent: Dry 50 g of Gooch grade long fiber asbestos at 302°F to 392°F (150°C to 200°C). Transfer to a dry 5-lb bottle and add 100 g of P_2O_5 powder, mixing it by rolling and shaking. Keep tightly closed. Pack the drying tube No. 17 under anhydrous conditions.

(b) Equipment, figure 1:

(1) Whitey needle valves, polytetrafluoroethylene packing, 1/4 inch Swagelok ends, or equivalent, for V_2, V_3, V_4, V_5, V_{18}, and V_{19}.

(2) Three each 1/4-inch tees, Swagelok ends, or equivalent.

(3) Compound gauge, 316 stainless steel diaphragm, 30 inches, vacuum to 100 psi pressure.

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(4) One-fourth inch copper tubing, cut to fit.

(5) Glass stopcocks, 3-way, 3 required for S_7, S_8, and S_9.

(6) One-fourth inch glass tubing, cut to fit.

(7) Polyethylene tubing, 1/4 inch I.D., cut to fit.

4.4.1.2.2 Assembly and passivation. The absorber manifold shall be prepared as follows: Disassemble each valve and fitting; flush each part including tubing with reagent grade dichloromethane and purge with an inert gas to dry. Assemble the apparatus as shown in figure 1, but do not make the glass-to-metal connection at V_{19}. The top of the reactor (No. 12) must be located above the absorber burette (No. 13). Connect the F_2 source, V_1 is closed. Close V_3 and V_{19}. Open V_2, V_4, V_5, V_6 and connect an argon source to V_{18}. Admit 50 psig of argon conforming to MIL-A-18455 to the metal system. Vent argon pressure by means of V_3 and V_{19}. Close V_3 and V_{19}. Repressurize with argon and vent in a similar manner three times to insure complete removal of air and moisture trapped in the manifold. Repressurize to 50 psig with argon. Check each valve, fitting, and weld for leaks with a soap solution. If no leaks are detected, prepare the apparatus for the analysis. If leaks are detected, remedy the situation and recheck for leaks. The argon pressure is released through V_3. Close V_3 and V_6. Passivate the apparatus by admitting approximately 10 psig F_2 slowly to the metal system at V_1 the F_2 sample location. Close V_1; allow system to passivate for 15 minutes. Connect the glass portion of the manifold to V_{19}. Orient stopcocks S_7, S_8, and S_9 so that a straight-through path is provided to scrubber No. 16. Open V_{19} slowly. As soon as the manifold reaches ambient pressure, the manifold may be used for F_2 sampling.

4.4.1.2.3 Fluorine sample. The sample for the non-condensables and the fluorine assay shall be obtained by the following procedure.

4.4.1.2.3.1 Procedure. After assembly and passivation of the manifold (figure 1, and paragraph 4.4.1.2.2) connect the F_2 sample source at V_2. Open V_2. Open the F_2 sample valve (V_1) slowly. Allow no more than 8 psig on the system. Do not allow the bubbling in the scrubbers No. 15 and 16 to be too vigorous. Purge the system with F_2 for 30 minutes. When all of the inert gases have been displaced, there will be little or no bubbling in scrubber No. 16. The reaction zone in reactor No. 12 is indicated by a "hot region" on the aluminum tube. When the location of the hot region indicates that more than two-thirds of the reactant has been consumed, the reactor should be recharged with NaCl. After the 30-minute purge with F_2, close V_{19} and orient S_9 so that all three side arms are connected. Then orient S_8 in the same fashion. Turn S_7 so that V_{19} and only the scrubber No. 13 are connected. Open V_{19} slowly. Control the F_2 flow with the sample valve (V_1). Collect 100 to 150 ml of gas in the top of absorber No. 13 when the liquid levels of No. 13 and 14 are equal. When the required amount of

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gas has been collected in No. 13, orient S_7, S_8, and S_9 to the purge position as before. Close the F_2 sample valve V_1. As soon as the bubbling in the scrubbers stops, open V_6 slowly to purge the manifold with argon. Close V_6, V_{19}, and V_{18}, and orient S_8 and S_9 to the vent position. The assay sample, as chloride ion, is contained in the liquid portions of reactors No. 13 and No. 14 while the gas phase above the liquid in No. 13 contains the non-condensable gases, measure the volume of these non-condensable gases and record the barometric pressure and temperature.

4.5 Test methods.

4.5.1 Non-condensables. The total moles (n_t) of non-condensables are calculated as follows:

$$n_t = \frac{V}{RT}\frac{(B-10)}{760}$$

Where:

B = barometric pressure in mm Hg at the time of the volume measurement of gas in the top of absorber 13, paragraph 4.4.1.2.3.1

T = temperature of the gas in degrees Kelvin.

V = volume of the gas in milliliters as measured in paragraph 4.4.1.2.3.1.

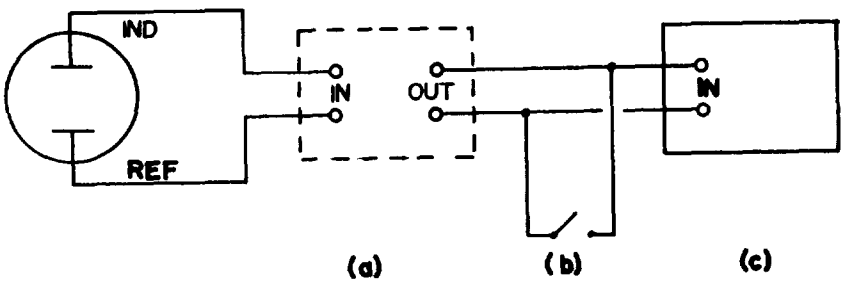
R = 82.06 $\frac{\text{ml atms}}{\text{mole degree K.}}$

The vapor pressure of H_2O above the absorber solution is approximately 10 mm Hg at ambient laboratory conditions. The total measured pressure must be corrected accordingly to obtain the partial pressure of non-condensables.

4.5.2 Fluorine assay. The F_2 content shall be determined by the following procedure.

4.5.2.1 Procedure. The reduced chlorine, contained in absorbers No. 13 and 14 is quantitatively transferred to a 2000-ml volumetric flask. The absorbers are washed five times with approximately 100 ml of water each time, and the washings combined in the 2000-ml flask. Add water to mark and mix by inverting. Allow to cool to room temperature before adding more water to the mark. Repeat until no further shrinkage in liquid volume occurs. (Caution: This solution is strongly alkaline, causes burns!) The assay is conducted potentiometrically, in duplicate, using sensing and suggested read-out equipment as shown in figure 2. Two 50-ml aliquots are withdrawn from the 2000-ml volumetric flask and placed in the titration beakers, each containing a polytetrafluorethylene coated stirring bar. Add 3 to 5 drops of phenolphthalein

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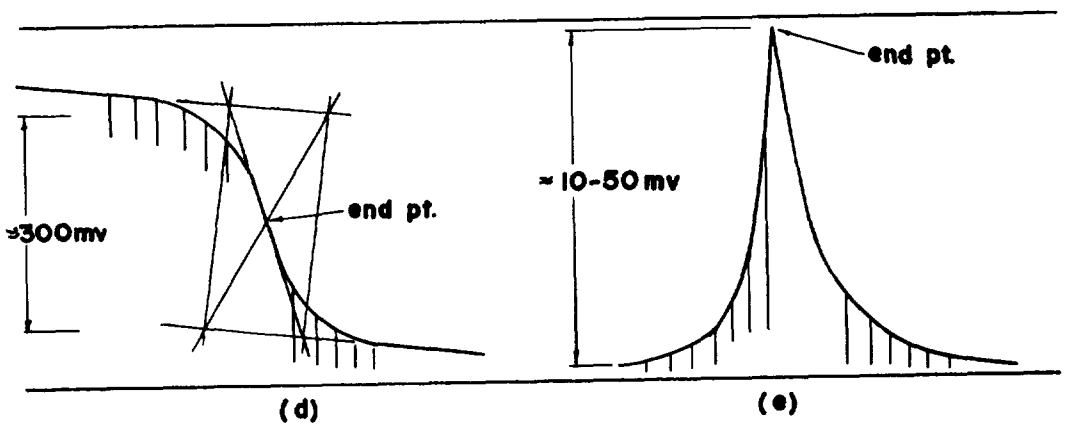
Ref = Reference electrode, saturated calomel.

Ind = Indicator electrode, silver/silver chloride billet.

(a) = Time derivative computer, optional, omit with pH meter.

(b) Bell button on-off switch, omit with pH meter.

(c) Potentiometer, strip chart recorder, or pH meter switched to "mv".



(d) Strip chart tracing, no derivate, or pH meter mv, when plotted vs volume, ml.

(e) Strip chart tracing of the derivative.

Figure 2. Suggested Potentiometric Apparatus

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indicator and neutralize the alkaline aliquot with concentrated HNO_3, stirring slowly until solutions are colorless. Add 1 ml concentrated HNO_3 in excess. Wash down the sides of each beaker with chloride-free water until each is at least half-filled (200-300 ml). Insert the Ag/AgCl-S.C.E. electrode assembly and titrate each aliquot with standardized 0.5 N $AgNO_3$ potentiometrically. Titrate 25 ml of alkaline potassium arsenite stock solution, treated in the same manner as the 50 ml aliquots of absorbers No. 13 and 14. This is the reagent blank. In all cases, it is necessary to check the electrical functioning of the apparatus prior to titrating the two aliquots. This may be accomplished when standardizing the 0.5 N $AgNO_3$ in triplicate, or by inserting the electrodes into solutions containing an excess Cl^-, rinsing the electrodes with water, and then inserting them into a solution with excess Ag^+.

4.5.2.2 Calculation. The total moles of fluorine shall be calculated as follows:

moles of F_2 (n_{F_2}) = A x N x 0.0200

Where:

A = (aliquot ml of $AgNO_3$-ml $AgNO_3$ reagent blank)

N = normality of $AgNO_3$

The precision of the replicate aliquot titrations should be within 0.1 percent.

4.5.2.3 Reagents and equipment. The following reagents and equipment shall apply as conditions of 4.5.2.

(a) Reagents:

(1) Silver nitrate solution, 0.5 N: Dissolve 84.945 g of ACS reagent grade silver nitrate in chloride-free distilled water in a red (actinic) 1000-ml volumetric flask and dilute to mark. Stopper and keep in a dark place. Standardize against "primary standard" grade, dried, sodium chloride, using the procedure of 4.5.2.1.

(2) Sodium chloride standard. Standard stock solution: Weigh out accurately to 0.1 mg approximately 30.0 g of sodium chloride and dissolve with chloride-free distilled water in a 1000-ml volumetric flask and dilute to the mark. Withdraw 3 each 25 ml aliquots with a volumetric pipet for standardization of the silver nitrate. Each 25-ml

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aliquot contains an accurately known number of milligrams (mg) of NaCl. Add 1 ml of concentrated HNO_3, an excess of H_2O, and titrate with $AgNO_3$ as per procedure of 4.5.2.1.

$$\text{Normality } AgNO_3 = \frac{25 \times W \times f}{58.456 \times a}$$

Where: W = mg NaCl per 1 ml.

a = ml $AgNO_3$ used to the end-point.

f = wt % NaCl assay (purity)/100.0. (i.e. 99% = 0.99)

(3) Chloride-free distilled or deionized water (test with $AgNO_3$).

(4) Nitric acid, ACS reagent grade.

(5) Silver nitrate, ACS reagent grade.

(6) Sodium chloride, primary standard grade. Dry at 230°F (100°C) overnight and cool in a desiccator.

(b) Equipment:

(1) Beakers, 400-600 ml capacity plus watch glasses.

(2) Magnetic stirring bars, polytetrafluoroethylene coated.

(3) Stirring motor.

(4) Electrode assembly, Ag/AgCl and saturated calomel (S.C.E.).

(5) 50 ml buret, buret stand.

(6) 25 ml, 50 ml volumetric pipets.

(7) 1000 ml volumetric flask, clear glass.

(8) 1000 ml volumetric flask, red (actinic) glass.

(9) Polyethylene "squeeze" bottle for distilled water.

(10) 2000 ml volumetric flask.

(11) Desiccator.

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4.5.3 HF and CO_2 determination. HF and CO_2 shall be determined by the following procedure.

4.5.3.1 Preparation. Assemble the calibration manifold in accordance with figure 3. Disassemble each valve, fitting, and line to be used in the calibration manifold and flush with reagent grade dichloromethane. Dry each part with an inert gas. Reassemble the manifold. Assemble the sample cell in accordance with figure 4. Prepare a carbon/KOH trap as shown in figure 5. Place this trap between V_5 and the vacuum pump. Construct a mercury manometer as shown in figure 6. Connect the manometer to V_4. Connect a 5 psig argon source to V_6. Connect a source of pure HF to V_2. Connect the 7.5 cm cell to V_3. Close V_8 and open V_2, V_3, V_4, V_5, and V_7 to evacuate the system. Close V_5 and note manometer reading. If the manometer reading does not change for 10 minutes, the system is free from leaks. If leaks are detected, remedy the situation before proceeding. If no leaks are detected, open V_1, the HF source, and admit approximately one atmosphere of HF to the system. Allow the HF to remain in the system for 15 minutes. Close V_1 and evacuate the system through V_5.

4.5.3.2 Calibration. Calibration curves for HF and CO_2 shall be obtained as follows. Refer to figure 3. Close V_5 and open V_1 slowly and admit 40 mm of HF to the 7.5 cm IR cell. Record the manometer reading (P_1). Close V_6, V_2, and V_7 and open V_5 to evacuate the manifold. Close V_5 and open V_6 slowly to admit argon to the manifold. When the argon approaches one atmosphere, open V_7 and allow the pressure to rise to one atmosphere. Close V_6, V_3, and V_7 and remove the cell. Determine the absorbance of the mixture in the cell, on a near-infrared spectrophotometer, scanning the region between 2300mu and 2600mu and measuring the most sensitive peak. Use a CaF_2 crystal blank in the reference beam to compensate for absorbance of the window material. Connect the cell to V_3 and open V_3 and V_5 to evacuate the manifold. Close V_5 and open V_7. Open V_5 and evacuate the manifold and cell to approximately 1/2 atmosphere. Close V_5 and record the manometer reading (P_2). Close V_7 and open V_5 to evacuate the manifold. Close V_5 and open V_6 slowly to admit argon to the manifold. When the pressure approaches one atmosphere open V_7 and allow the pressure to rise to one atmosphere. Close V_6, V_3, and V_7 and remove the 7.5 cm cell from the manifold. Determine the absorbance of the HF in Ar at the same wavelength previously used. Repeat this dilution and measurement procedure at least five times.

Calculate the HF in the cell for the first standard as follows:

$$\frac{(B-P_1)}{B} \times 100 = Q_1$$

Where B = barometric pressure, mm Hg.

P_1 = first manometer reading ($L_H - L_b$) = difference between high and low leg of manometer, mm Hg.

Q_1 = vol % HF in first standard.

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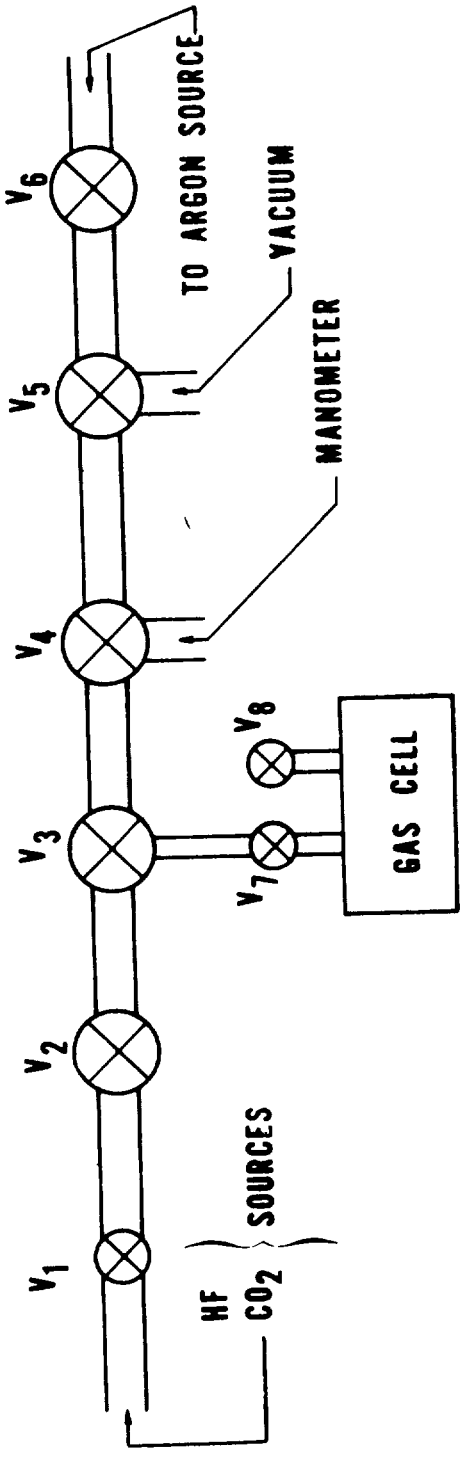


Figure 3. Calibration Manifold Apparatus

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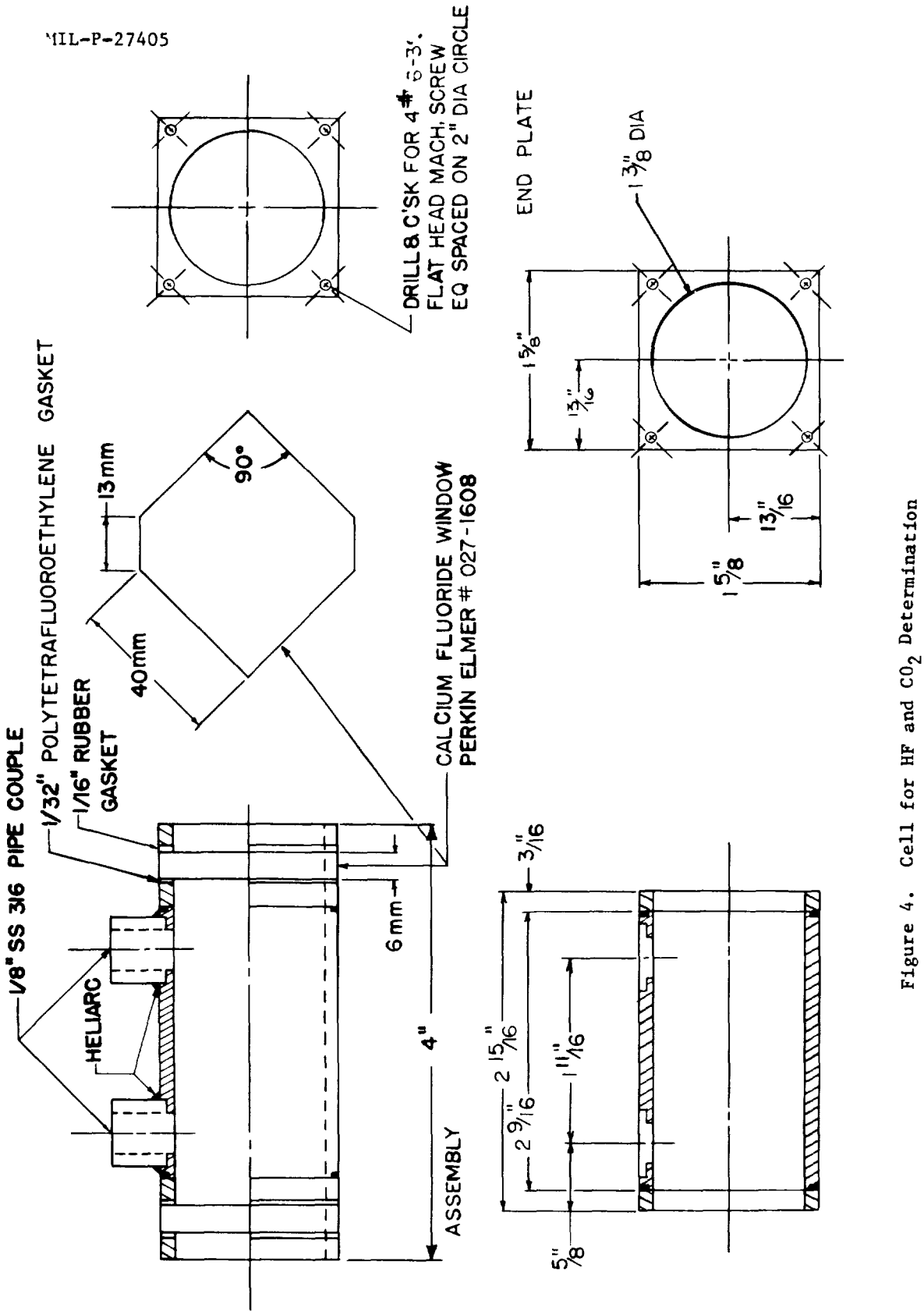


Figure 4. Cell for HF and CO_2 Determination

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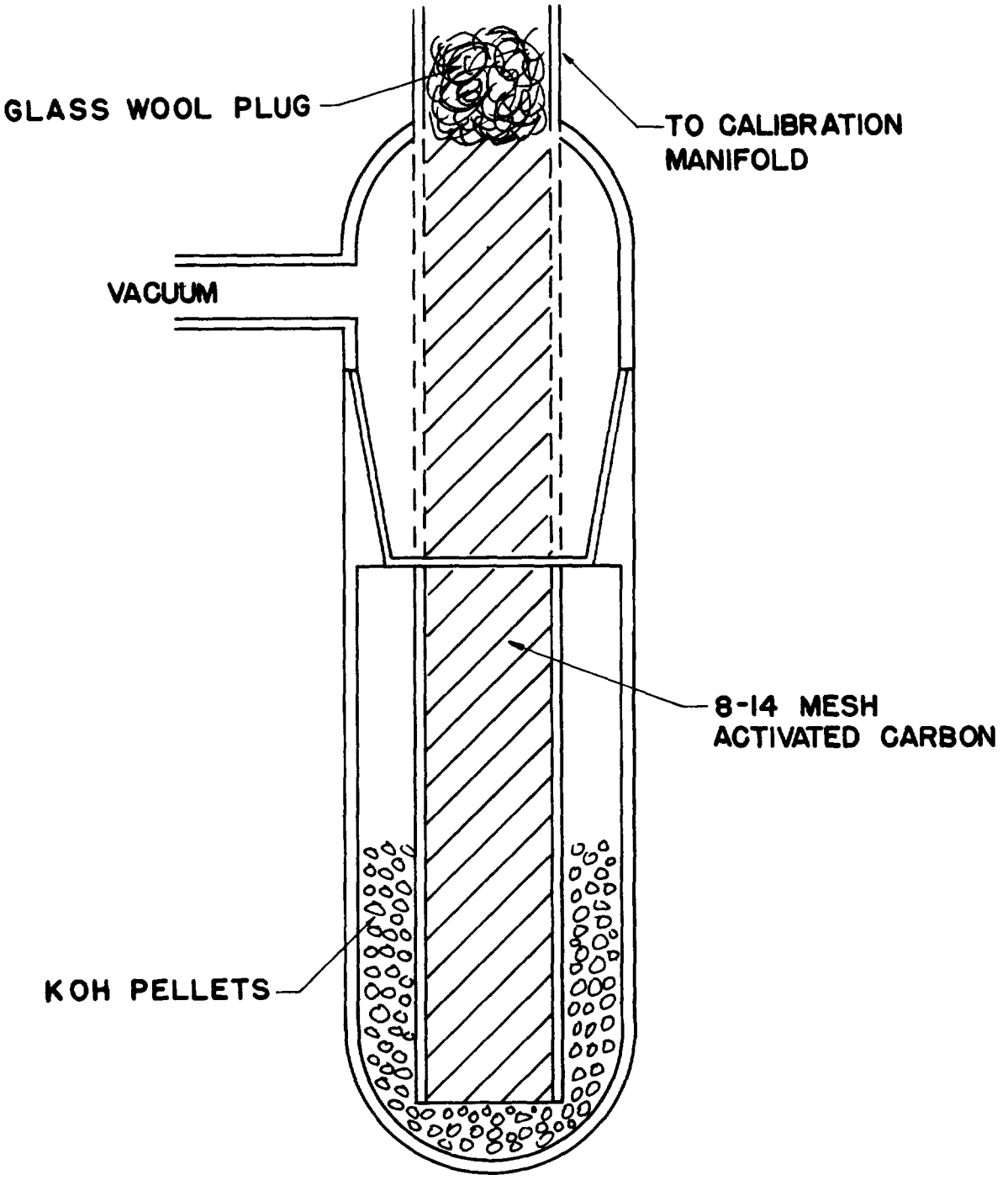


Figure 5. Carbon/KOH Trap

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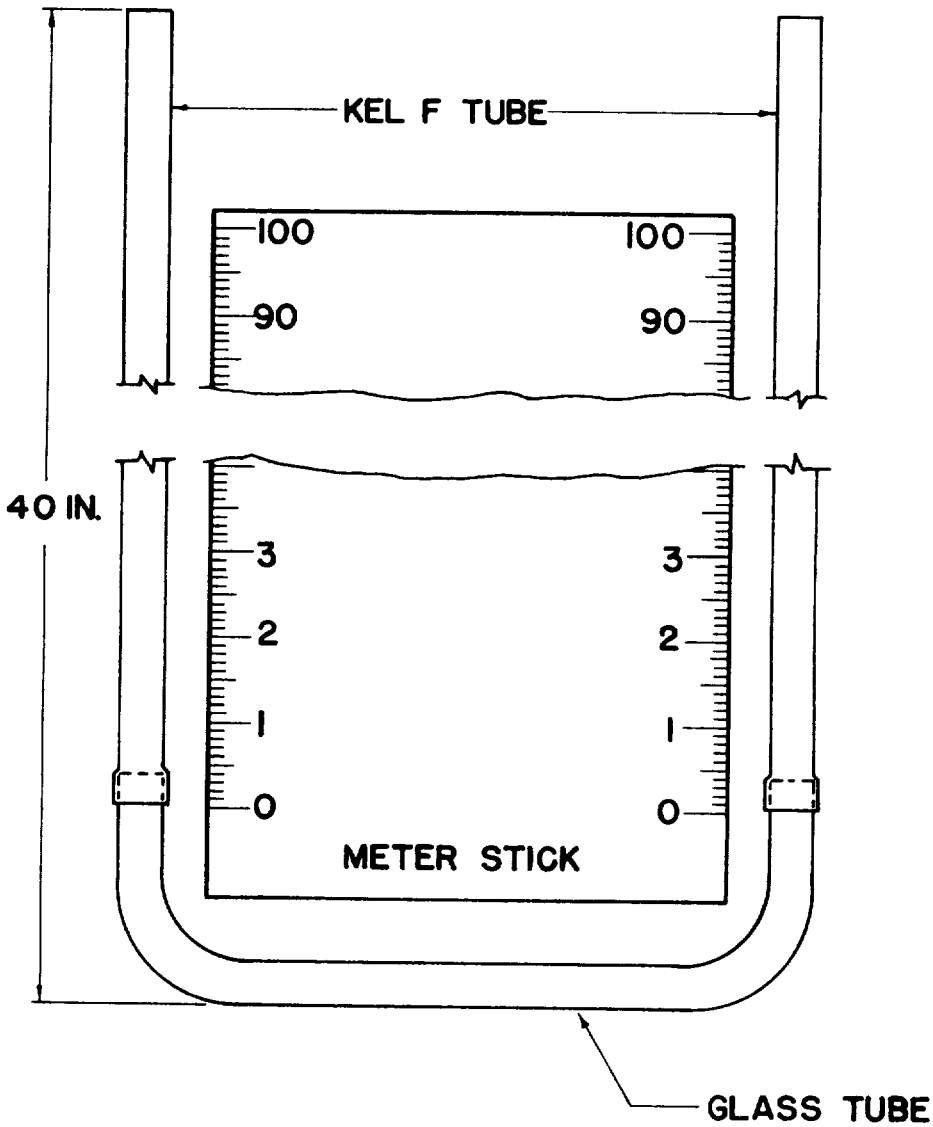


Figure 6. Open End Manometer

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The next five standards are calculated as follows:

$$\frac{(B-P_n)\ (Q_{n-1})}{B} = Q_n$$

Where: $P_n = P_2$ to P_6

$Q_{n-1} = Q_1$ to Q_5

Q_n = vol % HF in each subsequent dilution

Plot the absorbance of the standards versus the percent HF in the standards. Draw a curve through the points.

To calibrate for CO_2, repeat the HF procedure 4.5.3.2 with the following exceptions:

(a) Substitute a pure CO_2 source.

(b) Scan the 3.0μ to 5.0μ region with the infrared spectrophotometer.

(c) Measure the absorbance at 4.30μ.

(d) Plot $\log \frac{Io}{I}$ vs vol % CO_2.

4.5.3.3 Sample analysis. Attach the 7.5 cm cell to V_3 of the fluorine assay manifold (figure 1). Open V_3 and V_4. Open V_{20} and V_{21} of the cell. Close V_5 and V_2. Connect a cleaned length of copper tubing to V_{21} of the 7.5 cm cell so that the open end of the tube touches the flame of an air/gas burner. The burner flame will convert the F_2 to CF_4 and HF. When F_2 emerges from the open end of the tube, it will pop, and burn smoothly. Connect a 10 psig argon source to V_{18}. Open V_{18}, then open V_6 slowly. Purge the cell with argon for several minutes. Close V_6. Open V_2, then open the F_2 sample valve slowly and pass F_2 through the cell for 5 minutes. Close the F_2 sample valve and then close V_2 and V_3 and V_{20} and V_{21} of the cell. Remove the cell from the system. Use the 7.5 cm cell for both the HF and the CO_2 determinations. Use the CaF_2 crystal in the reference beam as in 4.5.3.2. Use the HF and CO_2 calibration curves prepared earlier for calculating the volume percent HF and CO_2 in the sample.

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4.5.3.4 Reagents and equipment. The following reagents and equipment shall apply as conditions of 4.5.3.

(a) Reagents

(1) Hydrogen fluoride (HF), Anhydrous, lecture bottle size.

(2) Carbon, activated 8/12 mesh, and potassium hydroxide, pellets.

(3) Carbon dioxide (CO_2), Bone dry grade, lecture bottle size.

(4) Argon, MIL-A-18455.

(b) Equipment

(1) Cell, special, 7.5 cm, constructed as shown in figure 4.

(2) Spectrophotometer, near infrared, Beckman DK-2, Cary 14, or equivalent.

(3) Spectrophotometer, infrared, Beckman IR-5, or equivalent.

(4) Valves, 2 ea, 1/8 inch pipe to 1/4 inch Swagelok ends. Whitey, or equivalent, for V_7 and V_8 on the 7.5 cm cell.

(5) Copper or stainless 1/4 inch tubing, cut to fit.

(6) Valve, 1/4 inch Swagelok, 316 Stainless steel, Whitey or equivalent, 2 required, V_1 and V_6.

(7) Valves, stainless steel bellows, 3-way, 1/4 inch tubing ends, "Hoke HTP" or equivalent, 3 required, V_3, V_4, and V_5. These valves are always open across the "T". They open or close the side arm only.

4.5.4 Final calculations. The final calculations shall be made using the following procedure.

4.5.4.1 Procedure. The volume percentages of F_2 and the non-condensables are calculated as follows:

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$$A = 100-(\text{vol \% HF} + \text{Vol \% } CO_2)$$

$$\text{Volume \% non-condensables} = \frac{A \cdot n_t}{n_{F_2} + n_t}$$

$$\text{Volume \% } F_2 = \frac{A \cdot n_{F_2}}{N_{F_2} + n_t}$$

Where:

n_{F_2} = moles of F_2 calculated in paragraph 4.5.2.2.

n_t = moles of non-condensables calculated in paragrpah 4.5.1.

4.6 Preparation for delivery inspection. The preservation, packaging, packing, and marking for shipment and storage of the propellant shall be inspected to determine compliance with the requirements of section 5 of this specification.

4.7 Rejection. When any sample of the propellant fails to conform to the requirements specified herein, the entire lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The propellant, gas or liquid, shall be packaged in cylinders, portable tanks, cargo tanks or tank cars as specified by the procuring activity. All containers shall comply with the requirements of the Department of Transportation (DoT), Regulations 49 CFR 171-190, or special permit obtained from DoT by the shipper in conjunction with Commander, Headquarters Military Traffic Management and Terminal Service, Washington, D. C. 20315.

5.1.1 Cylinders. Cylinders for gaseous fluorine shall conform to DoT Specifications 3A1000 or 3AA1000 carbon steel or 3BN nickel. Cylinder valves shall have protection caps. All outlet valves shall be constructed of compatible materials, have no safety relief device, and be tightly capped or plugged. The charge shall not exceed 400 psig at 70°F and the load shall not exceed 6 pounds of gas.

5.1.2 Other containers. Portable tanks, cargo tanks, or tank cars for gaseous or liquid fluorine shall conform to the requirements of DoT special permit.

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5.1.3 Liquid containers. All containers for liquid fluorine shall be of triple wall design. The inner area shall be for the lading, the intermediate area for the liquid nitrogen coolant and the outer area for the insulating material and vacuum. All construction materials shall be suitable for long term storage of the lading without degradation.

5.2 Labeling. Each container shall be labeled and placarded in accordance with DoT Regulations 49 CFR 171-190 or DoT special permit and the MCA Manual L-1.

5.2.1 Identification tag. In addition an identification tag impervious to climatic conditions shall be wired to the discharge valve of each container and shall contain the following information:

Propellant, _____

Specification MIL-P-27405

Type No. _____

FSN No. 9135-

Quantity _____

Cont. or Order No. _____

Name of Manufacturer _____

Name of Contractor (if different from mfg.)

Date of Manufacture _____

5.3 Markings. Markings for shipment other than specified in 5.2 shall be in accordance with MIL-STD-129. In addition, cylinders and containers shall be marked in accordance with MIL-STD-101 and MIL-STD-172.

6. NOTES

6.1 Intended use. The propellant covered by this specification is intended for use as an oxidizer in rocket engines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Method of shipment, type, and capacity of containers.

(c) Type of propellant.

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(d) Quantity by weight in pounds (avoirdupois).

(e) That two copies of the test report, signed by the contractor's representative, listing values obtained on all tests (qualitative values where method provides), should accompany each shipment delivered to the consignee. In addition, one copy should be furnished the AFRPL (RPPS), Edwards, California 93523.

6.3 Warning. To promote safety in the transportation of propellants in interstate commerce by motor vehicle, the shipper should assure that each driver possesses an MCA Chem-Card-Transportation Emergency Guide No. CC10. A complete manual of cards or the individual cards are available from the Manufacturing Chemists' Association, 1825 Connecticut Avenue, N.W., Washington, D. C. 20009.

6.4 Pollution control. U. S. Public Laws dictate increased effort to improve air, land, and water pollution control of toxic propellant vapors, leaks, spills, and disposal during all phases of manufacture, transfer, storage, and transportation operations. The manufacturer/supplier is enjoined to approach the appropriate pollution control district to mutually resolve all problem areas, and to develop adequate control and disposal methods for situations which are likely to develop in any of the phases.

Custodians:

Army - MI

Navy - AS

Air Force - 12

Review Activities:

Air Force - 19, 23, 68

Preparing Activity:

Air Force - 12

Civilian Agency Interest:

NAS

Project No. 9135-0014

Reviewer/user information is current as of the date of this document. For further coordination of changes to this document, draft circulation should be based on the information in the current DoD Index of Specifications and Standards.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-P-27405, Propellant, Fluorine

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-0270

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FOLD

AFRPL (RPPS)
EDWARDS, CALIFORNIA 93523

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UNITED STATES AIR FORCE
OFFICIAL BUSINESS

Director
Air Force Rocket Propulsion Laboratory (RPPS)
Edwards, California 93523

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343-218 (A-133)

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